

AUTOMATION INTERNSHIP
AN INTERNSHIP PROJECT REPORT

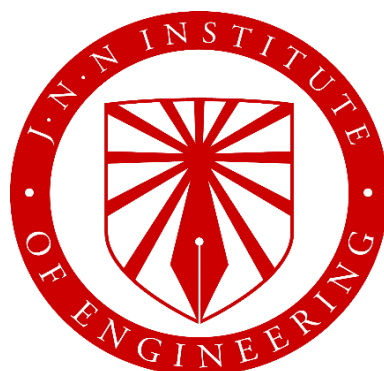
Submitted by REGAN MAYSAK.J

(Reg No:110725102206)

*in partial fulfillment for the award of the
degree of*

**BACHELOR OF TECHNOLOGY
IN**

Artificial Intelligence & Data Science



J.N.N Institute of Engineering
(An Autonomous Institute , Affiliated to Anna University, Chennai)

AUGUST – 2026 - 2027



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VISION OF THE INSTITUTE:

Lead the transformation of engineering and management learning experience to educate the next generation of innovators and entrepreneurs who want to make the world a better place.

MISSION OF THE INSTITUTE:

IM1: To develop the required resources and infrastructure and to establish a conducive ambience for the teaching-learning process.

IM2: To nurture professional and ethical values in the students and to instils in them a spirit of innovation and entrepreneurship.

IM3: To encourage a desire for higher learning and research in the students and to equip them to face global challenges.

IM4: To provide opportunities for students to learn job-relevant skills to make them industry ready.

IM5: To interact with industries and other organization to facilitate transfer of knowledge and know-how.



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Department of Artificial Intelligence & Data Science

VISION OF THE DEPARTMENT:

To impart quality-education, inculcate professionalism and enhance the problem-solving skills of the students in the domain of Artificial Intelligence & Data Science by applying recent technological tools and incorporating collaborative principles with a focus to make them industry ready.

MISSION OF THE DEPARTMENT:

- **DM 1:** To enhance the knowledge of the students with most recent advancements and refresh their insights in the field of Artificial Intelligence and Data Science.
- **DM 2:** To equip the students with strong fundamental concepts, analytical capability, programming and problem-solving skills.
- **DM 3:** To make the students industry ready and to enhance their employability through training, internships and real-time projects.
- 1. **DM 4:** To guide the students to perform research on Artificial Intelligence and Data Science, with the aim to provide solutions to the problems of the industry



PROGRAM OUTCOMES

S.NO	GRADUATE ATTRIBUTE	PROGRAM OUTCOMES
PO1	Engineering knowledge	Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis	Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/Development of solutions	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern Tool usage	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The Engineer and society	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice
PO7	Environment and sustainability	Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.



PO8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and team work	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM EDUCATIONAL OBJECTIVES (PEO)

S.No	PROGRAM EDUCATIONAL OBJECTIVES (PEO)
PEO1	To provide graduates with the proficiency to utilize the fundamental knowledge of basic sciences, mathematics, Artificial Intelligence, data science and statistics to build systems that require management and analysis of large volume of data.
PEO2	To enrich graduates with necessary technical skills to pursue pioneering research in the field of AI and Data Science and create disruptive and sustainable solutions for the welfare of ecosystems.
PEO3	To enable graduates to think logically, pursue lifelong learning and collaborate with an ethical attitude in a multidisciplinary team



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Department of Artificial Intelligence & Data Science

PROGRAM SPECIFIC OBJECTIVES (PSOs)

S.No	PROGRAM SPECIFIC OBJECTIVES (PSOs)
PSO1	Graduates should be able to evolve AI based efficient domain specific processes for effective decision making in several domains such as business and governance domains.
PSO2	Graduates should be able to arrive at actionable Fore sight, Insight , hind sight from data for solving business and engineering problems
PSO3	Graduates should be able to create, select and apply the theoretical knowledge of AI and Data Analytics along with practical industrial tools and techniques to manage and solve wicked societal problems.



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BONAFIDE CERTIFICATE

Certified that this internship project report entitled “AUTOMATION USING PYTHON AND REACT LIBRAIES” is the bonafide work of REGAN MAYSAK.J(Reg. No: 110725102206), who carried out the internship project work under the supervision of the Department of Artificial Intelligence and Data Science, J.N.N Institute of Engineering, Chennai.

Signature of the HOD with date

Dr. P. Nagarajan

Professor and Head,
Department of Artificial Intelligence
& Data Science Engineering,
**J.N.N Institute of Engineering,
chennai**

Signature of the Supervisor with date

MS.S. BRINDHA

Faculty Supervisor,
Department of Artificial Intelligence
& Data Science Engineering,
**J.N.N Institute of Engineering,
chennai**

Certified that the above students have attend of viva voce during the exam held on.....

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

First and foremost, we express our sincere gratitude to our Beloved **FOUNDER & CHAIRMAN Shri. S. JAYACHANDRAN B.Sc., B.L.**, and also heartfelt thanks to our **VICE CHAIRMAN Mr. NAVEEN JAYAHANDRAN M.S.**, for providing necessary facilities for the completion of our project.

We thank our **PRINCIPAL Dr. G. MOHANBABU, Ph.D.**, for the encouragement provided with the congenial environment essential for us to complete the project.

We take this opportunity to express our gratitude to our **HEAD OF THE DEPARTMENT Dr. P. NAGARAJAN** and all other faculty members in the Department of ARTIFICIAL INTELLIGENCE AND DATA SCIENCE for giving us continuous encouragement for completing our project.

I extend my sincere thanks to **MS.S. BRINDHA** for her guidance and support during the internship report preparation.

I express my heartfelt thanks to, Giona Technologies Private Limited, for providing me with the opportunity to undergo Automation Using Python And React Libraries from 03.08.2026 to 27.08.2026 and to work on the project titled “ SMART OFFICE CRM ”.

Finally, I express my heartfelt thanks to my parents and friends for their constant encouragement, motivation, and support throughout the completion of this internship and project report.

REGAN MAYSAK
(110725102206)

ABSTRACT

Automation plays an important role in improving the efficiency, consistency, and speed of software development and

application workflows. This report presents the work carried out during the internship at GIONA TECHNOLOGIEA PRIVATE LIMITED, from 1st August 2026 to 27th August 2026.

The main objective of this internship was to gain practical knowledge of automation using Python and React JS and to apply the concepts learned through hands-on implementation. Python was used for automation logic, scripting, and backend-oriented tasks, while React JS was used to develop interactive and responsive user interfaces.

The internship provided practical exposure to planning, developing, testing, and integrating software components. The work involved understanding application requirements, writing Python-based automation scripts, designing React components, and connecting application functionality in a structured workflow.

Throughout the internship, challenges related to program logic, component integration, debugging, and application behaviour were addressed through testing and continuous improvement. The experience strengthened my understanding of Python programming, React JS, automation concepts, coding practices, problem-solving, and independent learning.

Overall, this internship provided valuable practical exposure to automation using Python and React libraries and helped me improve my technical skills through hands-on development and implementation

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1. INTRODUCTION

1.1 OVERVIEW

Smart Office CRM is an automation-based application developed to simplify manpower management and task delegation in an organization. Managing employees, assigning tasks, monitoring work and maintaining task-related information manually can consume significant time and effort. The project uses **Python and React JS** to develop an organized system for managing manpower and assigning tasks. React JS is used to create the user interface, while Python is used for automation and backend-related processing.

1.2 PROBLEM STATEMENT

In a traditional office environment, manpower allocation and task delegation may be performed manually using spreadsheets, messages or other separate methods. This can make it difficult to maintain accurate information about employees and their assigned tasks.

1.3 OBJECTIVES

- To develop an automated office management application.
- To manage manpower information in an organized manner.
- To simplify the process of task delegation.
- To reduce repetitive manual work.
- To provide an easy-to-use user interface.
- To use Python for automation-related operations.
- To use React JS for developing the application interface.
- To improve task organization and monitoring.
- To reduce errors in manual task allocation.
- To provide a foundation for further office automation.

1.4 SCOPE OF THE PROJECT

The scope of Smart Office CRM includes automation of manpower and task delegation activities.

1.5 OVERVIEW OF THE PROJECT

The Smart Office CRM follows a simple workflow. The user accesses the application through the React JS interface. Manpower information and task details are entered into the system. The system processes the information and supports task delegation using Python-based automation.

2. LITERATURE REVIEW

2.1 INTRODUCTION TO RESUME ANALYSIS

The Smart Office CRM follows a simple workflow. The user accesses the application through the React JS interface. Manpower information and task details are entered into the system. The system processes the information and supports task delegation using Python-based automation.

2.2 AUTOMATION IN OFFICE MANAGEMENT

Office automation refers to the use of software and computing technologies to perform repetitive office activities with minimum manual intervention.

2.3 MANPOWER MANAGEMENT USING AUTOMATION

In a manual system, employee details and work assignments may be maintained using spreadsheets or documents. Managing such information becomes difficult when the number of employees and tasks increases.

2.4 TASK DELEGATION USING AUTOMATION

In the proposed system, task information can be entered into the application and processed for assignment. Python-based automation can be used to perform repetitive task-related operations.

2.5 TOOLS AND TECHNOLOGIES USED

The main technologies used in the project are:

Python

Python is used for automation and processing. Its simple syntax and availability of libraries make it suitable for developing automation applications.

React JS

React JS is used to develop the frontend user interface. It allows the application to be divided into reusable components and provides a responsive user experience.

Smart Office CRM

The developed application combines Python-based automation with a React JS interface to manage manpower and task delegation.

3. METHODOLOGY

3.1 OVERALL APPROACH

The project development follows a systematic approach.

The major steps are:

1. Identify the office management problem.
2. Collect the required requirements.
3. Design the application workflow.
4. Develop the React JS interface.
5. Implement Python-based automation.
6. Develop manpower management functionality.
7. Develop task delegation functionality.
8. Test the system.
9. Identify and correct errors.
10. Evaluate the completed application.

3.2 REQUIREMENT DATA COLLECTION

The first stage of the project is requirement collection.

The requirements are identified based on the need for:

- Manpower management.
- Task creation.
- Task assignment.
- Task tracking.
- Automated processing.
- User-friendly interface.

3.3 MANPOWER DATA MANAGEMENT

Manpower data management is an important part of the Smart Office CRM.

The system can maintain information required for identifying employees and assigning work.

The manpower management process includes:

Manpower Information → Data Entry → Processing → Storage/Management → Task Assignment

Organized manpower information makes it easier to identify suitable employees for different tasks.

3.4 TASK CREATION AND DELEGATION

The task delegation process begins with creating a task.

The basic process is:

Create Task → Enter Task Details → Select Manpower → Assign Task → Record Task → Monitor

The system provides an organized way to manage the task assignment process.

Automation reduces the amount of repetitive manual work involved in this process

3.5 DATA PROCESSING

The Data processing involves receiving the information entered through the application and performing the required operations.

Python can be used for processing task and manpower-related information.

The general processing flow is:

Input Data → Validation → Processing → Automation → Output

Proper processing helps maintain consistency in the application

3.6 AUTOMATION WORKFLOW

The The automation workflow is one of the main components of the project.

The workflow can be represented as:

User Input → React Interface → Data Processing → Python Automation → Task Operation → Result

Python performs the required automation operations, while React JS provides the interface through which the user interacts with the system.

3.7 SYSTEM WORKFLOW

The complete Smart Office CRM workflow can be represented as:

USER



REACT JS USER INTERFACE



MANPOWER / TASK INFORMATION



PYTHON PROCESSING



AUTOMATION



TASK DELEGATION



TASK MANAGEMENT / OUTPUT

This workflow provides a connection between the frontend interface and automation process.

4. IMPLEMENTATION

4.1 PYTHON ENVIRONMENT AND LIBRARIES

Python is used as one of the main technologies in the Smart Office CRM project.

Python is suitable for automation because of its simple syntax and availability of libraries for data processing and application development.

The Python environment is used to implement the automation logic required for manpower and task management.

Python performs processing operations based on the requirements of the application.

4.2 REACT JS ENVIRONMENT AND LIBRARIES

The React JS is used for developing the frontend of the Smart Office CRM application.

React allows the application interface to be developed using reusable components.

The interface can contain components such as:

- Dashboard.
- Manpower section.
- Task section.
- Task assignment section.
- Status section.
- Navigation components.

React JS provides an interactive interface for users.

4.3 USER INTERFACE IMPLEMENTATION

The user interface is designed to make the application simple and easy to use.

The interface provides access to different operations of the Smart Office CRM.

The main interface can include:

- Dashboard.
- Employee/manpower information.
- Task details.
- Task assignment.
- Task status.
- Navigation options.

React JS is used to implement these interface components.

4.4 MANPOWER MANAGEMENT IMPLEMENTATION

The manpower management module is used to organize employee-related information.

The implementation provides a structured method for handling manpower information required for task delegation.

The basic process is:

Add Manpower → Store Information → View Manpower → Use for Task Assignment

This module supports the main objective of the project, which is automation of manpower management.

4.5 TASK DELEGATION IMPLEMENTATION

The task delegation module allows tasks to be created and assigned to manpower.

The user can provide the required task information and select the appropriate manpower.

The system processes the information and records the task assignment.

This reduces the need for repetitive manual task allocation.

4.6 PYTHON-BASED AUTOMATION

Python is used to implement the automation part of the project.

The automation process handles repetitive operations related to manpower and task delegation.

The basic concept is:

Input



Python Processing



Automation Logic



Task Operation



Output

Using automation helps make the workflow more systematic and reduces manual effort.

4.7 SMART OFFICE CRM IMPLEMENTATION

The Smart Office CRM integrates the major project components into one application.

The application combines:

- React JS frontend.
- Python-based automation.
- Manpower management.
- Task management.
- Task delegation.

The integration of these components creates an organized office management system.

5. RESULTS AND DISCUSSION

5.1 USER INTERFACE RESULTS

The developed user interface provides access to the major functions of the Smart Office CRM.
The React JS interface allows users to interact with manpower and task-related features.
The interface is designed to provide clear navigation and organized information.

5.2 MANPOWER MANAGEMENT RESULTS

The manpower management functionality provides a structured method for handling employee information.
The system makes it easier to identify available manpower and use the information during task delegation.
This supports better organization of manpower-related activities.

5.3 TASK DELEGATION RESULTS

The task delegation functionality provides an organized process for assigning work.
The user can create a task, provide the required information and assign it to suitable manpower.

5.4 AUTOMATION RESULTS

Python-based automation helps perform repetitive operations related to task and manpower management.
The automation workflow reduces unnecessary manual intervention and provides a systematic process.
The result is a more organized approach to office task delegation.

5.5 SMART OFFICE CRM RESULTS

The Smart Office CRM successfully combines the frontend and automation components into one project structure.

The major results include:

- Organized manpower management.
- Structured task delegation.
- Reduced repetitive manual operations.
- User-friendly interface.
- Python-based automation.
- React JS-based frontend.

The project demonstrates how automation can be applied to office management activities.

5.6 SYSTEM PERFORMANCE ANALYSIS

System performance can be evaluated based on:

- Response time.
- Task processing.

- User interface responsiveness.
- Automation efficiency.
- Data handling.
- Accuracy of task assignment.

Testing the application with different manpower and task inputs helps identify errors and improve the system. The performance of the system can be further improved by optimizing the automation logic and application components.

6. CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The **Smart Office CRM** project demonstrates the use of automation for manpower management and task delegation.

The project uses **Python and React JS** to create an organized office management application. React JS provides the user interface, while Python supports the automation and processing requirements.

The project helped provide practical knowledge of automation, frontend development, application workflow and task management.

Overall, the project demonstrates how repetitive office activities can be organized and automated using modern programming technologies.

6.2 LIMITATIONS

The current system may have certain limitations.

- The functionality depends on the implemented automation logic.
- Advanced analytics may not be available.
- Additional modules may be required for complete CRM functionality.
- Large-scale deployment may require further optimization.
- More testing is required with different real-world office scenarios.

6.3 FUTURE ENHANCEMENTS

The project can be improved by adding:

1. Employee attendance management.
2. Automated notifications.
3. Email integration.
4. Task priority management.
5. Deadline reminders.
6. Employee performance tracking.
7. Automated report generation.
8. Data analytics and dashboards.
9. Role-based login.
10. Advanced task allocation.

These features can make the application more useful for larger organizations.

6.4 FUTURE SCOPE

The future scope of Smart Office CRM is broad because office activities contain many repetitive operations that can be automated.

The project can be extended into a complete office management platform with features such as employee management, attendance, task tracking, performance analysis, notifications and reporting.

Python-based automation can also be expanded to perform more intelligent operations, while React JS can be used to develop a more advanced and interactive interface

INTERNSHIP CERTIFICATE



Old No. 66, New No. 39, Thirumala Pillai Road, T. Nagar, Chennai - 600017, Tamil Nadu

Date: 27 August 2026

INTERNSHIP COMPLETION CERTIFICATE

TO WHOMSOEVER IT MAY CONCERN

This is to certify that **Regan Maysak J** (Reg No.: 110725102206) has successfully completed a Part-Time Internship at Giona Technologies Private Limited from 01 Aug to 26 Aug 2026, for a duration of 26 days, in the capacity of an Executive Intern in the Automation domain.

The internship training was provided by Faculty Mr. Steffano Charles, who guided the intern through the assigned training activities and practical exposure relevant to the Automation domain.

The HR coordination for the internship was handled by Sriram V, HR Coordinator, Giona Technologies Private Limited.

During the internship period, the intern was involved in automation-related activities and responsibilities assigned as part of the internship programme. The intern demonstrated a sincere approach towards the work allotted, adhered to the assigned timelines, and showed a keen interest in learning and developing practical skills in the domain.

We wish the intern success in all future academic and professional endeavours.



For Giona Technologies Private Ltd

A handwritten signature in blue ink, appearing to read 'Ryherstef', written over a horizontal line.

Managing Director